

A Mental Health Platform for African Communities

By

NGEYEN VICTORY YINKFU

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September 2023

DECLARATION

I **Ngeyen Victory** do declare that this dissertation is my own work. I have to the best of my knowledge acknowledged all authors or sources from where I got information. I further declare that this work has not been submitted in any university or institution for the award of a degree or any of its equivalents.

APPROVAL

This research proposal is my original work and has not been presented for a degree in any other university.

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Student name: NGEYEN VICTORY

Signature: Date: / /2022

DECLARATION BY SUPERVISOR

This research proposal has been submitted with our approval as the university supervisor

Supervisor's name: ISHIMWE KEVIN OLIVIER ANGEL

Signature: Date..... /...../ 2023

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DEDICATION

I would like to dedicate this research dissertation to my family and friends that have been supporting me throughout my education journey.

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LIST OF ABBREVIATIONS

MH – Mental Health

MHD – Mental Health Directory

Whogotalk – Platform for communication between users and MH professionals

ASGI – Asynchronous Server Gateway Interface

JSX – JavaScript XML

CSS3 – Cascading Stylesheet

HTTPS – Hyper Text Transfer Protocol Secure

WS – Web sockets

REDIS – Remote Dictionary Server

TCP – Transmission Control Protocol

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ABSTRACT

Mental health is the condition that influences our minds in our daily activities. It determines how we handle stress, how we relate to others and how we make our choices. WHO famously defines health as “a state of mental well-being that enables people to cope with the stresses of life, realize their abilities, learn well and work well, and contribute to their community. (WHO, 2022)” Mental health is therefore integral to health and intimately connected with physical health and behavior. While there are efforts toward sensitization of mental wellness from a few non-governmental agencies in Africa, traditional societal beliefs and stigmas hamper the efforts of these organizations as topics around mental wellness are considered taboo in many communities. In this project, we propose a platform that links users with doctors and volunteers. We also link to a mental health Directory a knowledge repository about mental health to help our volunteers better assist users who might want to learn more about mental illnesses to properly support people going through mental health issues. MHD was developed through the application of information retrieval theories to source and aggregate content from verified online sources. Whogotalk is a platform that encourages to MHD to provide live assistance to people who need help. The system will be developed with the aid of Visual Studio Code, Django, Redis Server, ASGI, Web sockets, React. The languages that will be used to develop the system include Python, JavaScript, JSX, CSS

1. CHAPTER ONE: GENERAL INTRODUCTION

1.1. Introduction

This chapter covers the background of the study, statement of the problem, objectives of the study, specific objectives, scope of the study, the significance, and limitations of the study.

1.2. Background of the Study

"Mental health is an integral component of overall health and well-being" (World Health Organization). However, mental health challenges are often overlooked or stigmatized, particularly in low- and middle-income countries. According to the WHO, mental health disorders account for 10% of the global burden of disease, and in Africa, mental health conditions are estimated to account for 19% of the total years lived with disability (WHO, 2018). In Africa, there is a growing concern about mental health, with an increasing number of individuals experiencing mental health challenges such as anxiety, depression, and post-traumatic stress disorder (PTSD) (Betancourt, et al., 2013). Despite this, access to mental health services in Africa is often limited, and even when services are available, they may not be affordable or culturally appropriate (NIH, 2019). Furthermore, mental health topics are often considered taboo in many African communities, making it difficult for individuals to seek (Collins, et al., 2011).

Access to mental health services is a critical issue in many African communities, particularly in rural and remote areas (Collins, et al., 2011). According to the World Health Organization (WHO), there is a significant gap in mental health services in Africa, with many people unable to access the care they need. This is due to a range of factors, including a shortage of mental health professionals, inadequate funding for mental health services, and limited infrastructure in many areas (Collins, et al., 2011).

In addition to these challenges, there is also significant stigma associated with mental health issues in many African communities (NIH, 2019). Mental health issues are often considered taboo, and people who suffer from mental health conditions are often stigmatized and excluded from their communities (NIH, 2019). This stigma can lead to a reluctance to seek help, further exacerbating the issue of limited access to mental health services.

The consequences of inadequate access to mental health services and the stigma associated with mental health issues are significant. Mental health conditions can have a profound impact on a person's quality of life, affecting their ability to work, socialize, and carry out day-to-day activities. In addition, untreated mental health conditions can lead to other health problems, such as substance abuse and physical illness

(Betancourt, et al., 2013). Despite these challenges, there is growing recognition of the importance of addressing mental health in Africa, and initiatives have been launched to improve access to mental health services and to reduce stigma and discrimination (Collins, et al., 2011). This project aims to contribute to these efforts by creating a platform that provides 24/7 access to mental health support and guidance from licensed therapists and trained volunteers. By providing a safe and confidential space for people to access mental health resources, this project hopes to reduce the stigma associated with mental health issues in African communities and to improve access to care for those who need it most.

1.3. Statement of the Problem

Mental health is a crucial component of overall well-being, yet mental illnesses continue to be stigmatized and neglected in many African communities (Gureje, et al., 2005). This stigma often prevents individuals from seeking professional help or discussing their issues with family and friends, leading to a lack of access to necessary mental health services (Chibanda, et al., 2016). Furthermore, there is a shortage of mental health professionals in many African countries, with only one psychiatrist per 1 million people in some areas (World Health Organization, 2020). This scarcity exacerbates the issue, making it difficult for those in need to receive adequate treatment and care.

In Rwanda, mental health issues are prevalent, with depression affecting up to 15% of the population (Munyandamutsa, et al., 2018). Despite this, there is a significant lack of mental health resources and infrastructure in the country (Rwanda Biomedical Centre, 2018). This lack of resources is especially detrimental in rural areas where access to mental health services is limited (Nsereko, et al., 2013). As a result, mental health issues are often left untreated, leading to further negative impacts on individuals and society as a whole.

Therefore, there is a critical need for accessible and effective mental health services in Rwanda. The development of a mental health platform that links users with trained professionals and volunteers could be a viable solution. By utilizing technology, this platform could overcome barriers to access and provide support to those in need, while also educating the community on mental health issues and reducing the associated stigma (Fakoya, et al., 2019). This study aims to investigate the feasibility and potential effectiveness of such a platform in improving mental health outcomes in Rwanda.

1.4. Objectives of the Project

1.4.1. General Objectives

The general objective of this project is to create a platform that connects individuals in African communities with licensed therapists and trained volunteers, providing 24/7 access to mental health support and guidance. The platform will aim to reduce the stigma associated with mental health issues in African communities and increase access to mental health resources.

1.4.2. Specific Objectives

The specific objectives of the project are:

- Develop a user-friendly platform for mental health education and support.
- Create a secure login and registration system for users, doctors, and volunteers.
- Allow users to search for and connect with volunteer mental health experts for live chat sessions.
- Enable users to search for mental health information through an integrated knowledge repository.
- Provide an appointment scheduling system for users to book virtual consultations with mental health professionals.

By achieving these specific objectives, this project aims to make a significant contribution to improving mental health services in African communities and addressing the significant challenges posed by limited access to care and stigma associated with mental health issues.

1.5. Research Questions

To guide the project and achieve its objectives, the following research questions will be addressed:

- What are the barriers to accessing mental health services in African communities?
- What are the common mental health issues affecting individuals in African communities?
- What are the best practices for providing mental health support and guidance via online platforms?
- How can technology be leveraged to reduce the stigma associated with mental health issues in African communities?
- How can the platform be designed to ensure accessibility, security, and confidentiality for individuals seeking mental health support and guidance?
- How can awareness and education on mental health issues be promoted in African communities through the platform?

By answering these research questions, the project can develop a platform that meets the needs of African communities and effectively addresses the challenges posed by limited access to mental health services and the stigma associated with mental health issues.

1.6. Scope of the Study

The study will focus on developing a messaging platform that connects users with mental health professionals and trained volunteers in a culturally appropriate and sensitive way. The platform will be designed to allow users to chat with mental health professionals and volunteers in a secure and confidential manner. The study will also recruit mental health professionals and volunteers who have experience working with African communities to provide support through the platform.

1.6.1. Content Scope

The platform will offer mental health support and guidance to individuals in African communities. The content will cover a wide range of mental health issues, including anxiety, depression, trauma, grief, and substance abuse. The platform will provide access to licensed therapists and trained volunteers who can offer support and guidance to individuals seeking help. In addition, the platform will offer information and resources on mental health issues, self-help tips, coping strategies, and information on available mental health services.

1.6.2. Geographical Scope

The platform will be accessible to individuals in African communities. While the project aims to reach as many individuals as possible, it will initially target specific countries in Africa. The target countries will be identified based on factors such as the prevalence of mental health issues and the availability of mental health services. The platform will be accessible via the internet, and individuals will be able to access the platform using their smartphones or computers. An offline version will be available as a MHD.

1.6.3. Time Scope

The project ran for a period of 4-6 months. During this period, the researcher developed the platform. After the launch of the platform, the researcher will assess the impact of the platform and determine whether to continue or expand the project.

1.7. Significance of the Project

This study is significant because it addresses a critical need for accessible and culturally appropriate mental health services in Africa. The project is significant in the following ways:

1.7.1. Personal Interest

The project is of personal interest to the individuals in African communities who are seeking mental health support and guidance. By providing access to licensed therapists and trained volunteers, the platform can help individuals who are struggling with mental health issues to receive the support they need to improve their mental health and well-being. The platform can also help to reduce the stigma associated with mental health issues, which can encourage individuals to seek help without fear of judgment or discrimination.

1.7.2. Institutional Interest

The project is of institutional interest to mental health organizations and healthcare providers in African communities. By leveraging technology to provide mental health support and guidance, the platform can help to address the challenges posed by limited access to mental health services in African communities. The platform can also help to improve the efficiency and effectiveness of mental health services by providing a platform for licensed therapists and trained volunteers to offer their services to individuals in need.

1.7.3. Public Interest

The project is of public interest to African communities and society at large. Mental health issues are prevalent in African communities, and the lack of access to mental health services can have a significant impact on individuals, families, and society as a whole. By providing a platform for mental health support and guidance, the project can help to improve the mental health and well-being of individuals in African communities. In addition, by promoting awareness and education on mental health issues, the project can help to reduce the stigma associated with mental health issues and promote greater understanding and acceptance of mental health as a vital component of overall health and well-being.

1.8. Limitations of the Study

The study has several limitations, including a limited geographic scope and a focus on a specific population. The study may also face challenges in recruiting mental health professionals and volunteers who have experience working with African communities to provide support through the platform. Additionally, the study may be limited in its ability to evaluate the long-term effectiveness of the platform in addressing mental health challenges in African communities.

1.9. Organization of the Project.

The project is organized as follows,

Chapter 1: In this chapter, which bears the name general introduction and background of the study, it covers; introduction and background of the study, problem statement, specific objectives, general objective, and limitations of the study.

Chapter 2: This chapter basically looks at the Literature review, empirical review, conceptual framework, research gaps, and summary.

Chapter 3: In this Chapter called research methodology the researcher will look at the Research Methodology:

Chapter 4: This chapter entitled Systems, Analysis and Design looks at the System Analysis and Design and will cover DFD diagram, ERD diagram, Database Schema, Data Dictionary.

Chapter 5: Conclusion and Recommendations looks at the conclusion and recommendations.

2. CHAPTER TWO: LITERATURE REVIEW

2.1. Introduction

Mental health issues are prevalent in African communities, but the lack of access to mental health services and the stigma associated with mental health topics can make it difficult for individuals to seek help. Technology has the potential to address some of these challenges by providing a platform for individuals to receive mental health support and guidance. In this chapter, we will review the existing literature on the use of technology in mental health services in Africa, including the benefits, challenges, and potential limitations of such platforms.

2.2. Definition of Key Concepts

2.2.1. Mental Health

Mental health includes our emotional, psychological, and social well-being. It affects how we think, feel, and act (CDC, 2023). It includes the ability to cope with life's challenges, build positive relationships, and function effectively in daily life.

2.2.2. Stigma

In relation to mental illness, stigma is when someone is marked or discredited somehow, or reduced from being a whole person to being a stereotype or labelled as a collection of symptoms or a diagnosis (Health Direct Australia, 2021). Stigma can prevent individuals from seeking help and can lead to discrimination and social exclusion.

2.2.3. Websockets (WS)

WebSocket is a communications protocol for a persistent, bi-directional, full duplex TCP connection from a user's web browser to a server. A WebSocket connection is initiated by sending a WebSocket handshake request from a browser's HTTP connection to a server to upgrade the connection. (TechTarget, 2016)

2.2.4. Redis Server

Redis, which stands for Remote Dictionary Server, is a fast, open source, in-memory, key-value data store. The project started when Salvatore Sanfilippo, the original developer of Redis, wanted to improve the scalability of his Italian startup. (AWS - Amazon, n.d.)

2.2.5. Asynchronous Server Gateway Interface (ASGI)

2.3. Review of Past Studies

2.3.1. Empirical Review

Several studies have investigated the use of technology in mental health services in Africa. A study by (Lund, et al., 2018) found that teletherapy was an effective way to increase access to care and improve treatment outcomes in Kenya. The study involved the use of video conferencing to deliver mental health services to patients in rural areas who would not have had access to care otherwise. The results showed significant improvements in symptoms of depression and anxiety among the patients who received teletherapy.

Another study by (Naslund, et al., 2017) examined the use of text messaging to deliver a mental health intervention in Kenya. The study found that the intervention was effective in reducing symptoms of depression and anxiety, as well as increasing treatment engagement and satisfaction among participants.

In South Africa, a study by (Kigozi, et al., 2019) investigated the use of mobile health technology to deliver mental health services to adolescents. The study involved the use of a mobile app to provide cognitive-behavioral therapy to adolescents with depression. The results showed significant reductions in depressive symptoms among the participants who received the intervention.

Overall, these studies demonstrate the potential of technology-based interventions to improve access to mental health services and to provide effective treatment to individuals in African communities. However, more research is needed to fully understand the benefits and limitations of such interventions, and to address the cultural and contextual factors that may impact their effectiveness.

The following platforms have been shown to be effective in providing mental health support and have become increasingly popular in recent years. However, they are mostly available to people with internet access and can be expensive for some users.

BetterHelp

This is an online therapy platform that connects users with licensed therapists. Users can choose to communicate with their therapist through text, phone, or video chat.

7 Cups

This is an online emotional support platform that offers free anonymous chat with trained listeners and therapists. They also have a paid service that connects users with licensed therapists.

Talkspace

This is an online therapy platform that connects users with licensed therapists. Users can communicate with their therapist through text, video chat, or voice message.

Ginger

This is a virtual mental health platform that provides on-demand access to behavioral health coaching, therapy, and psychiatry. Users can communicate with their therapist or coach through text, video chat, or voice message.

Woebot

This is a chatbot-based therapy platform that provides cognitive behavioral therapy (CBT) to users through a chat interface.

2.3.1.1. Existing Platforms

As discussed earlier, there are several existing platforms that provide mental health support through digital means. These platforms have been successful in reaching people who may not have access to traditional in-person mental health services. However, they can be expensive for some users and may not be available in all countries or regions. In addition, some users may prefer to communicate with mental health professionals in their local language, which may not be possible with existing platforms.

2.3.1.2. Proposed Platform

The proposed platform aims to address some of the limitations of existing platforms by providing a free, 24/7 chat-based service that connects users with mental health professionals and trained volunteers. The platform is designed with the African context in mind, taking into account the stigma associated with mental health topics in many communities in Africa. The platform also prioritizes providing mental health services in local languages, which may improve accessibility for some users. The platform will be open to anyone with an internet connection, making it accessible to people in areas where traditional mental health services may be unavailable or too expensive.

2.3.2. Summary of the Gap

While there are several existing platforms that provide mental health support through digital means, there are still gaps in the literature that need to be addressed. For example, there is a lack of research on the effectiveness of digital mental health services in low- and middle-income countries, particularly in the African context. There is also a need for more research on the barriers and facilitators to accessing digital mental health services in these settings. In addition, while existing platforms may provide services in multiple languages, there may still be a need for more tailored and culturally sensitive mental health services. Finally, there is a need for more research on the role of trained volunteers in digital mental health services, particularly in settings where mental health professionals may be scarce. Addressing these gaps will help to ensure that digital mental health services are effective, accessible, and culturally appropriate for users in low- and middle-income countries.

The project uses software like Adobe Photoshop CS4 for the creation of the images, Figma for Prototyping, Visual Studio Code as an Integrated development Environment, PostgreSQL for the database, Apache as the web server to serve static files, Redis Server for ASGI, Django for the Backend and React JS.

2.3.3. Conceptual framework

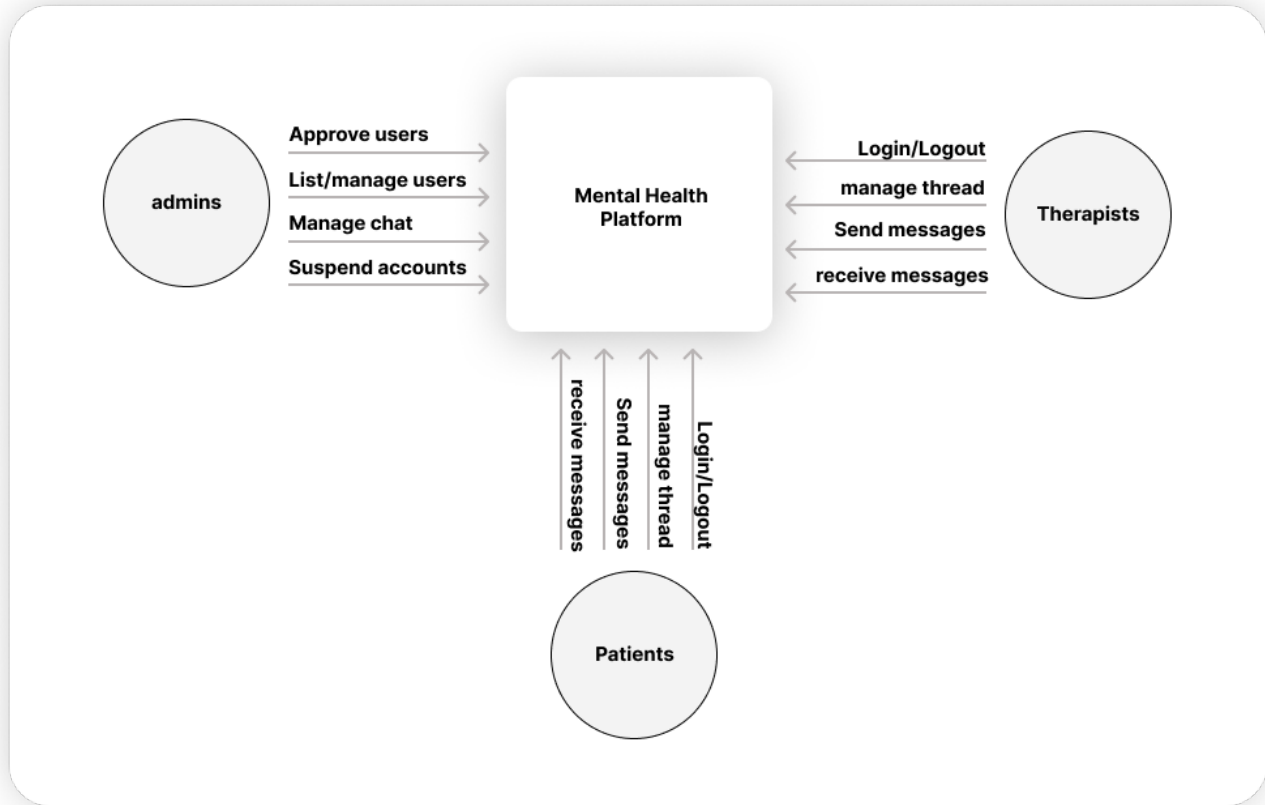


FIGURE 1: CONCEPTUAL FRAMEWORK

(Source: Researcher, 2023)

2.4. Conclusion

This literature review reviewed the related studies, empirical review, conceptual framework, and summary of the gaps. The researcher was able to research and obtain literature related to the study and present it in this chapter.

3. CHAPTER 3: RESEARCH DESIGN AND METHODOLOGY

3.1. Introduction

In this chapter, the researcher covers the research design, tools for data collection, the process of data collection, data analysis and presentation, reliability, and validity as well as the ethical consideration.

3.2. Research Design

The research design refers to the overall strategy that you choose to integrate the different components of the study in a coherent and logical way, thereby, ensuring you will effectively address the research problem; it constitutes the blueprint for the collection, measurement, and analysis of data. (De Vaus,2006)

This study will employ a quantitative research design.

3.2.1. Software engineering methods

Software engineering methods are just different ways of approaching software development and delivery.

Waterfall Model

According to the Institute of Electrical and Electronics Engineers (IEEE), software engineering means applying the principles of engineering to the software development field.

The Waterfall Model was the first Process Model to be introduced. It is also referred to as a linear-sequential life cycle model. It is very simple to understand and use. In a waterfall model, each phase must be completed before the next phase can begin and there is no overlapping in the phases. (Jayathilaka, 2020)

The waterfall Model illustrates the software development process in a linear sequential flow. This means that any phase in the development process begins only if the previous phase is complete. In this waterfall model, the phases do not overlap. Waterfall approach was first SDLC Model to be used widely in Software Engineering to ensure success of the project. In “The Waterfall” approach, the whole process of software development is divided into separate phases. In this Waterfall model, typically, the outcome of one phase acts as the input for the next phase sequentially.

- **Requirement Gathering and analysis** – All possible requirements of the system to be developed are captured in this phase and documented in a requirement specification document.
- **System Design** – The requirement specifications from the first phase are studied in this phase and the system design is prepared. This system design helps in specifying hardware and system requirements and helps in defining the overall system architecture.

- **Implementation** – With inputs from the system design, the system is first developed in small programs called units, which are integrated into the next phase. Each unit is developed and tested for its functionality, which is referred to as Unit Testing.
- **Integration and Testing** – All the units developed in the implementation phase are integrated into a system after testing of each unit. Post integration of the entire system is tested for any faults and failures.
- **Deployment of system** – Once the functional and non-functional testing is done; the product is deployed in the customer environment or released into the market.
- **Maintenance** – There are some issues that come up in the client environment. To fix those issues, patches are released. Also to enhance the product some better versions are released. Maintenance is done to deliver these changes in the customer environment.

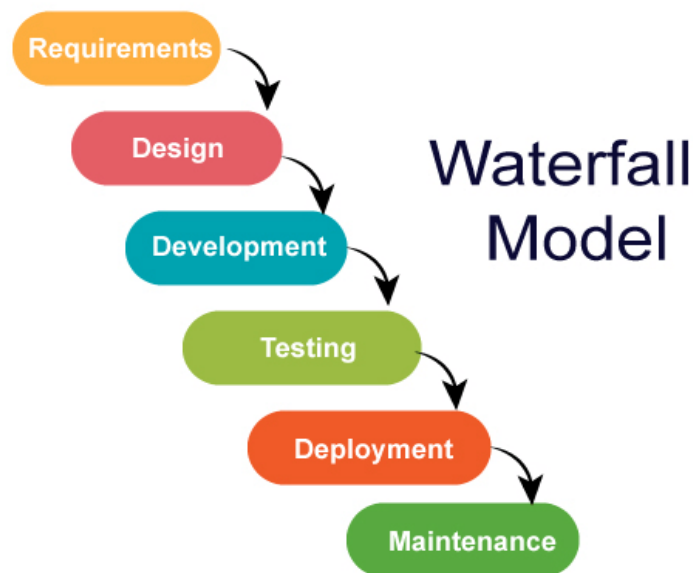


FIGURE 2: WATER FALL MODEL (MEDIUM.COM)

3.3. Population and Selection of the Sample

The population for this study consists of individuals who may require mental health support and assistance. Due to the practical limitations of reaching the entire population, a sample will be selected for the study. The selection of the sample will be done using a combination of probability and non-probability sampling techniques to ensure representation and diversity within the sample. The population of this study amounted to 20.

Sampling

In this qualitative case study, the researcher employed a random sampling method for participant selection, following the recommendation of (Johnson & Christensen, 2020). This approach was chosen due to the large number of potential respondents, ensuring a fair and unbiased representation. Sample size was calculated using Slovin's formula,

TABLE 1: CALCULATION OF SAMPLE SIZE

$n = \frac{N}{1 + Ne^2}$	Where, $n = \text{Sample Size}$ $N = \text{Population}$ $e = \text{margin of error}$
	Therefore, $n = \frac{21}{1 + 21(0.08)^2}$ $n = 18.51$

The sample size for this study is calculated to be 19.

3.4. Tools for Data Collection

To gather valuable data for the study, a combination of tools and methods specific to the mental health platform was utilized. We talk about these tools and methods in this section

3.4.1. Documentation on Mental Health Resources

Existing documents and resources related to mental health were thoroughly reviewed to gather information that would aid in the study. These documents included research papers, reports, and published materials from reputable sources. The documentation process aimed to enhance the knowledge base of mental health resources available to users on the platform.

3.4.2. Internet Research for Mental Health Information

This method aimed to gather up-to-date and relevant information on mental health conditions, treatment approaches, coping strategies, and available support services. The internet research process was crucial in ensuring that the mental health platform provides accurate and comprehensive information to its users.

3.4.3. Interviews with Mental Health Professionals and Users

Structured interviews were conducted with mental health professionals and platform users to gain insights into their experiences, challenges, and perspectives related to mental health. This approach aligns with the recommendation of (Kvale & Brinkmann, 2015) regarding the use of interviews in qualitative research.

3.4.4. Observation

Observation is way of gathering data by watching behavior, events, or noting physical characteristics in their natural setting. The researcher made overt observations, where participants were aware of being observed, to gather valuable insights. This aligns with the approach suggested by (Creswell & Poth, 2017) for conducting observational studies.

3.5. Data Collection Methods for the Mental Health Platform

To collect data on the mental health platform, a combination of quantitative and qualitative methods was employed. This approach aimed to gather both statistical data and rich, nuanced insights from users, ensuring a comprehensive understanding of the platform's impact and effectiveness. Questionnaires were also used as a means of advertising the study, as proposed by (Dillman, et al., 2014).

3.6. Data Analysis and Presentation

Data analysis involves descriptive and inferential analysis. Descriptive analysis focuses on presenting raw data, while inferential analysis establishes relationships between variables and tests for significance. The recommendations of Miles, Huberman, and Saldana (2019) were followed to effectively analyze and present the collected data.

3.7. Validity and Reliability

Validity and reliability were considered throughout the research process. To ensure validity, the research design and data collection instruments were carefully developed based on the principles outlined by (Lincoln & Guba, 1985). Reliability was addressed by using standardized data collection procedures and maintaining consistency in the application of research techniques.

3.8. Ethical Considerations

Ethical standards were strictly followed during the research process. The researcher obtained the necessary ethical clearance from the relevant institution, as advocated by (Babbie, 2016). Informed consent was obtained from all participants, and confidentiality and anonymity were maintained to protect the privacy and Identity of the respondents, in accordance with the guidelines provided by the American Psychological Association.

4. CHAPTER 4: SYSTEM ANALYSIS, DESIGN, AND IMPLEMENTATION

4.1. Introduction

This chapter presents an overview of the system analysis, design, and implementation process for the mental health platform. It outlines the key objectives, methodologies, and techniques employed to develop an effective and user-friendly platform that addresses the needs of individuals seeking mental health support. The chapter provides a comprehensive understanding of the system development life cycle, from initial requirements gathering to the final implementation and testing phases.

4.2. Data analysis and presentation

In this section, we discuss the data analysis and presentation techniques employed to derive meaningful insights from the data collected on the mental health platform. The analysis process aims to uncover patterns, trends, and correlations that can contribute to a better understanding of users' mental health needs and preferences. By applying appropriate statistical and qualitative analysis methods, the collected data is transformed into valuable information that guides decision-making and platform enhancements.

4.3. Interpretation of findings

From the findings obtained, the researcher interpreted them in the following ways.

Response Rate

From the 21 questionnaires handed out, 15 questionnaires were returned to the researcher. This represents 71.4% response rate

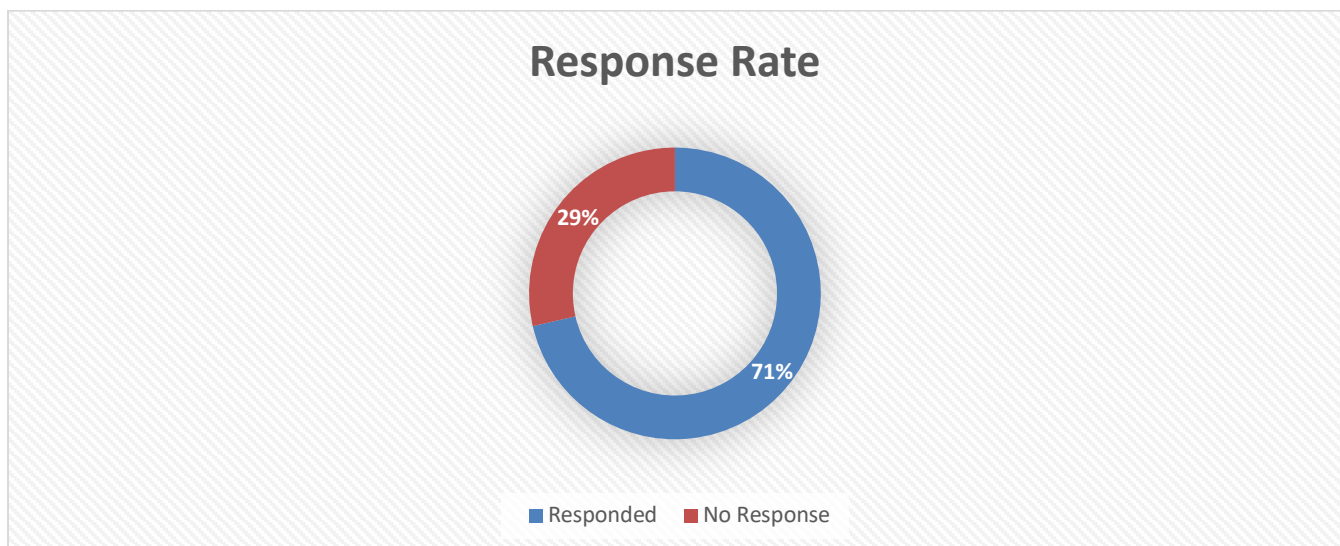


FIGURE 3: RESPONSE RATE

Current State of Mental health among Respondents.

Based on the responses gathered, here are the key findings regarding the state of respondents' mental health:

- **Average Mental Health:** About 22.2% of the respondents described their mental health as average. This suggests that they perceive their mental well-being to be neither particularly good nor poor.
- **Excellent Mental Health:** Approximately 11.6% of the respondents rated their mental health as excellent. This indicates a high level of well-being and positive mental state among this group.
- **Poor Mental Health:** Around 26.4% of the respondents reported having poor mental health. This suggests that they are facing significant challenges and may require additional support and resources.
- **Uncertainty about Mental Health:** About 10.7% of the respondents were unsure about the state of their mental health. This uncertainty may indicate a need for self-reflection or further evaluation to better understand their well-being.
- **Somewhat Good Mental Health:** Approximately 29.1% of the respondents stated that their mental health was somewhat good. This suggests that they experience a moderate level of well-being, with room for improvement.

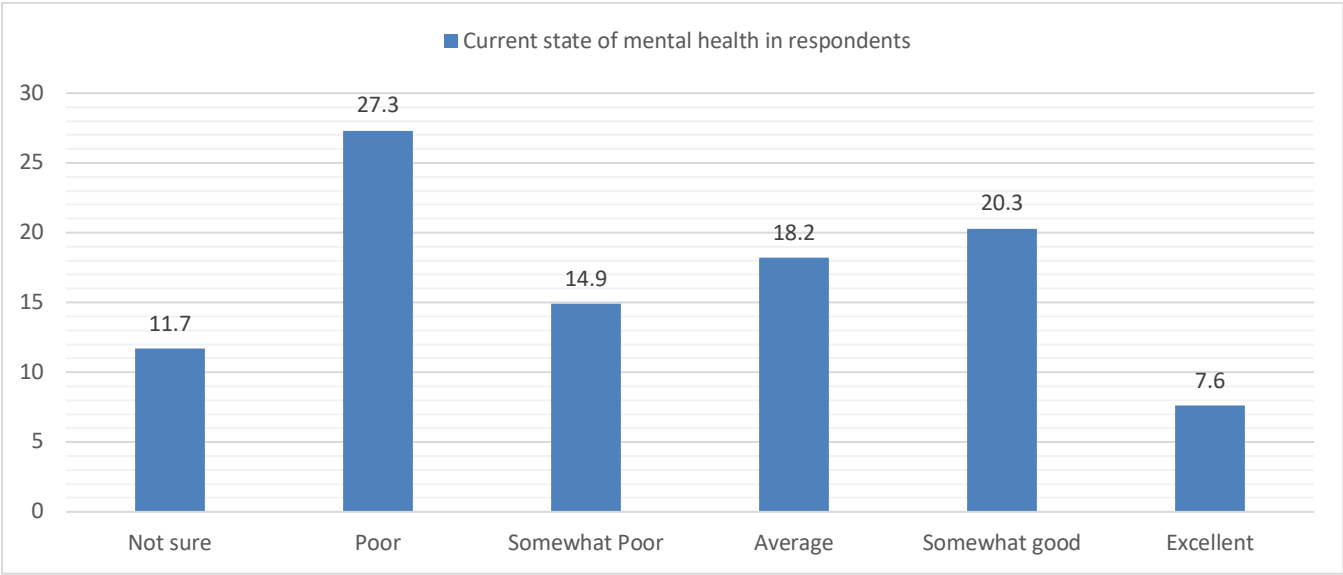


FIGURE 4: CURRENT STATE OF MENTAL HEALTH OF RESPONDENTS

4.4. Summary of Findings

These ratings highlight the diverse range of mental health experiences reported by the respondents. While a significant portion of respondents rated their mental health positively (excellent or somewhat good),

there is also a notable percentage who indicated poorer mental health or uncertainty about their mental health status.

By leveraging these findings, the mental health platform can tailor its services, resources, and interventions to cater to the specific requirements of different user segments. It can focus on enhancing mental well-being, providing coping strategies, and ensuring access to professional support for users with diverse mental health conditions and concerns.

4.5. Description of Existing System/Operations

In this section, we provide a description of the existing system or operations related to mental health support in the targeted communities. It is important to note that these communities currently lack access to formal mental health services, and individuals facing mental health challenges often face social stigma and exclusion.

The existing system in these communities is characterized by a significant absence of mental health support infrastructure. Limited resources, including mental health professionals, counseling centers, and support networks, contribute to a lack of accessible and comprehensive services. As a result, individuals grappling with mental health issues often find themselves isolated, unable to seek appropriate help or guidance.

4.6. Description of the new system

This project is composed of three (3) different modules, i.e.

Administrators, Patients and Therapists

1. Administrators

The administrator through this portal will be able to perform the following functions.

- Approve therapists and support staff
- View List of users and their roles
- Manage chats.
- Search users
- Suspend user accounts.
- Fix issues with user interactions.

2. Patients

The Patients through the chat application will be able to perform the following functions.

- View all therapists that are online.
- Setup user accounts.
- Chat with therapist
- View thread
- Delete thread
- Explore mental Health directory

3. Therapist

Through the cat application, therapists will be able to perform the following functions.

- View thread
- Account Setup
- Delete thread
- Engage patients
- Explore Mental health directory

4.7. Illustration of New system

4.7.1. Data Flow Diagram and processes

Context Diagram

A context diagram provides a high-level view of a system or process, illustrating the interactions between the system and its external entities. In the context of the mental health platform, a context diagram can visually represent the relationships and interactions between the platform and various stakeholders or external entities.

The mental health platform serves as the central system and is represented at the center of the context diagram. Surrounding the platform are the key external entities or stakeholders that interact with the system.

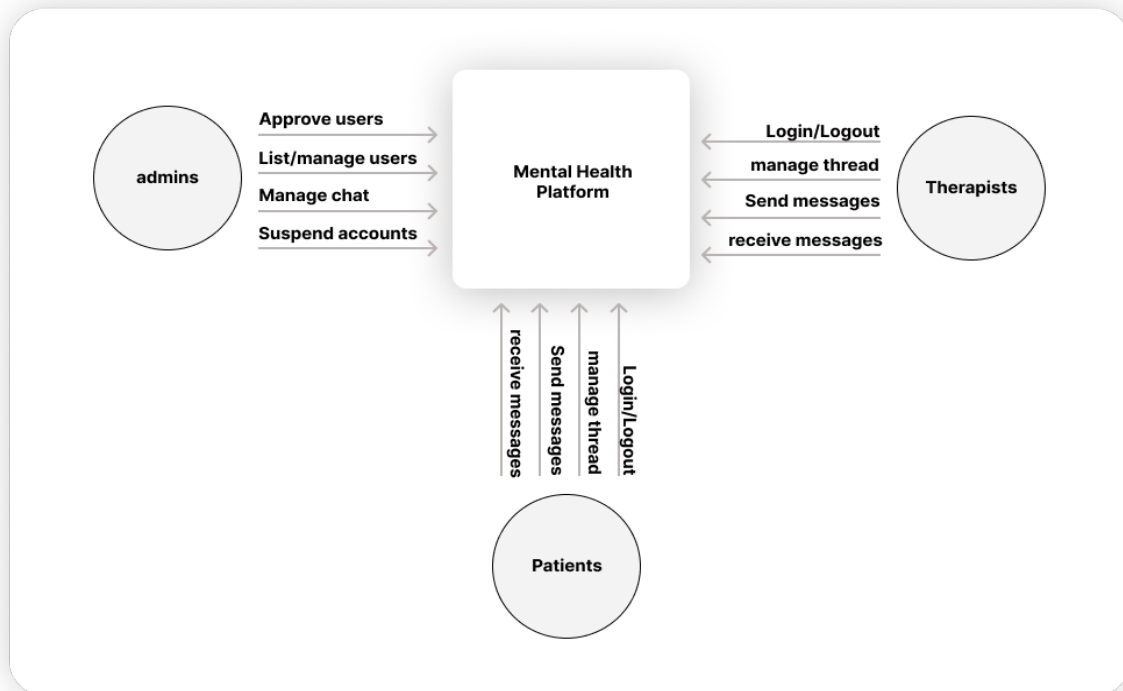


FIGURE 5: CONTEXT DIAGRAM

(Source: Researcher, 2023)

Data Flow Diagram Level 1

A Level 1 Data Flow Diagram (DFD) provides a more detailed representation of the data flow within a system by decomposing the processes identified in the Context Diagram. It shows the main processes and data flows between them, providing a clearer understanding of how data is processed and transformed within the system.

It's important to note that this Level 1 DFD provides an overview of the main processes and data flows within the mental health platform. It may not capture every detail or sub-process within each process. However, it serves as a useful visual representation to understand the flow of data and interactions between different components of the system.

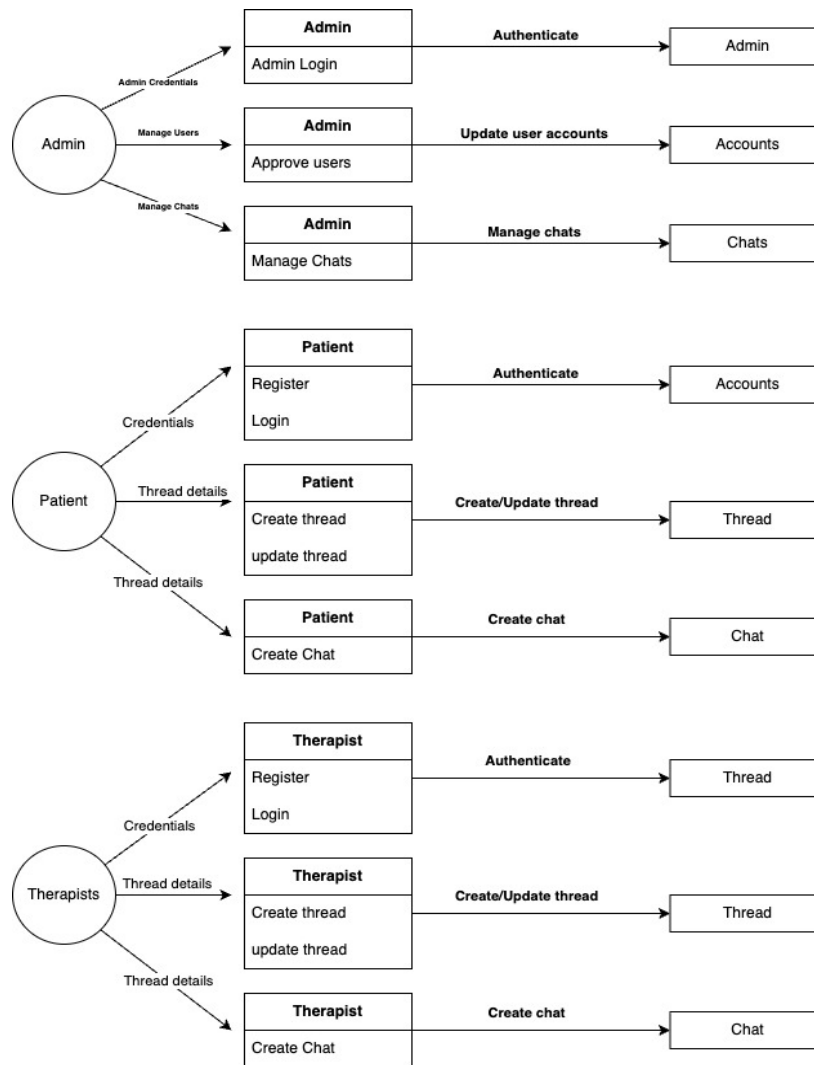


FIGURE 6: DATA FLOW DIAGRAM LEVEL 1

(Source: Researcher, 2023)

4.7.2. Use Case and Sequence Diagrams

Use Case and Sequence Diagrams are valuable tools in system analysis and design that help to illustrate the functionality and interaction of a system's components. In the context of the mental health platform, these diagrams can provide a visual representation of how different actors and components interact with each other.

Both the Use Case Diagram and Sequence Diagram are valuable tools in system analysis and design. They help stakeholders and developers gain a better understanding of the system's functionality, its interactions with users and other actors, and the overall flow of activities within the system. These diagrams aid in identifying requirements, clarifying system behavior, and ensuring that the final implementation aligns with the intended functionality of the mental health platform.

- The purpose is to show the interactions between the use case and actor.
- To represent the system requirements from the user's perspective.
- An actor could be the end-user of the system or an external system.

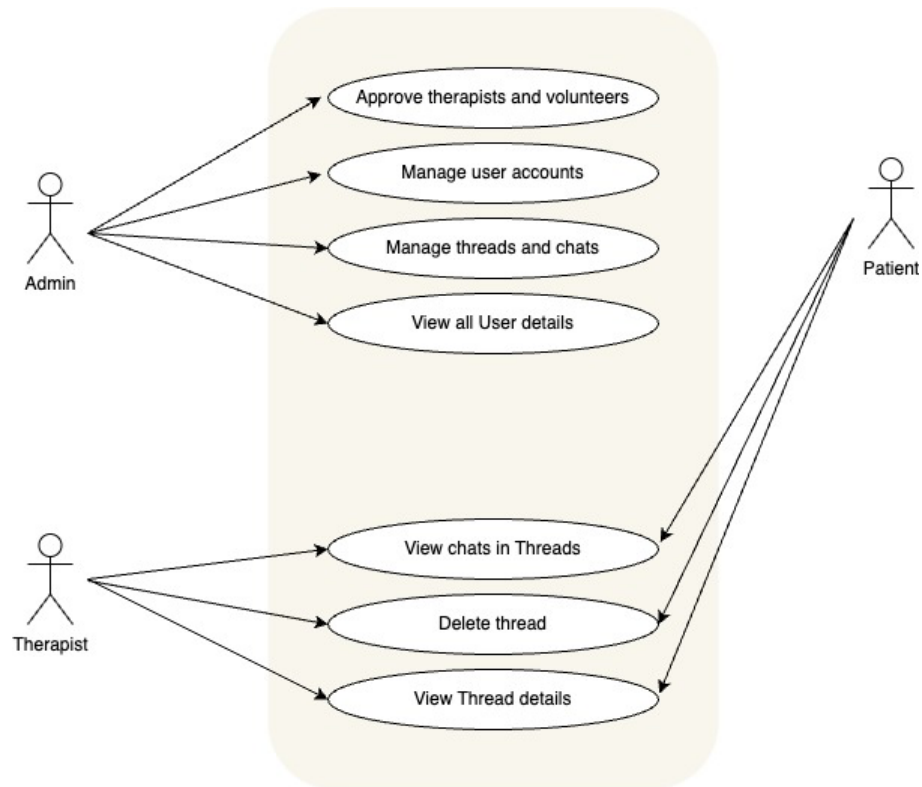


FIGURE 7: USE CASE DIAGRAM

(Source: Researcher, 2023)

4.7.3. Database Normalization

Database normalization is a process used in database design to eliminate data redundancy and ensure data integrity and consistency. It involves organizing the data into multiple related tables and applying normalization rules to minimize data duplication. The following are the commonly recognized normalization rules:

Normalization Rule

Normalization rules are divided into the following normal forms:

- First Normal Form
- Second Normal Form
- Third Normal Form
- BCNF

- Fourth Normal Form

First Normal Form (1NF):

For a table to be in the First Normal Form, it should follow the following rules:

- Eliminate repeating groups or arrays of values within a single row.
- Ensure each column holds atomic (indivisible) values.
- Store data in a tabular format with unique column names.

Second Normal Form (2NF):

For a table to be in the Second Normal Form, it should follow the following rules:

- Fully depend non-key columns on the entire primary key.
- Separate data into multiple tables to eliminate partial dependencies.
- Identify and establish relationships between tables using foreign keys.

Third Normal Form (3NF):

For a table to be in the Third Normal Form, it should follow the following rules:

- Remove transitive dependencies by ensuring non-key columns depend only on the primary key.
- Analyze and eliminate redundant data by splitting tables based on functional dependencies.
- Create separate tables for related entities to maintain data integrity.

4.7.4. Data Dictionary

TABLE 2: USERS DATA DICTIONARY TABLE

Column	Type	Null	Default	Links to	Comments	Media type
Id (<i>Primary</i>)	int(11)	No				
username	varchar (50)	No				
Full_name	varchar(100)	No				
Is_superuser		No	False			
Is_staff		No	False			
Is_admin		No	False			
Is_active		No	True			
Last_login	timestamp		current_timestamp()			
Date_joined	timestamp		current_timestamp()			

(Source: Researcher, 2023)

TABLE 3: USER PROFILES DATA DICTIONARY TABLE

Column	Type	Null	Default	Links to	Comments	Media type
Id (<i>Primary</i>)	int(11)	No				
user	Int	No		User		
role	varchar (50)	No				
Is_therapist	boolean	No				
Date Updated	timestamp		current_timestamp()			
Date Created	timestamp		current_timestamp()			

(Source: Researcher, 2023)**TABLE 4: CHAT ROOMS DATA DICTIONARY TABLE**

Column	Type	Null	Default	Links to	Comments	Media type
Id (<i>Primary</i>)	int(11)	No				
first	int	No		User		
second	Int	No		<u>User</u>		
Timestamp	timestamp		current_timestamp()			

(Source: Researcher, 2023)**TABLE 5: : MESSAGES DATA DICTIONARY TABLE**

Column	Type	Null	Default	Links to	Comments	Media type
Id (<i>Primary</i>)	int(11)	No				
Content	varchar (50)	No				
Sender	int(11)	No		User		
Recipient	Int(11)	No		User		
Room	Int(11)	No		ChatRoom		
sent_at	datetime	No	Timestamp			
Read_at	datetime	No	False			

(Source: Researcher, 2023)

4.7.5. Entity Relationship Diagrams (ERDs)

Entity Relationship Diagrams (ERDs) are commonly used in database design and serve as a visual representation of the relationships between entities. According to (Kroenke & Auer, 2016), ERDs provide a graphical illustration of the entities, attributes, and their associations within a system or database. They help in capturing the logical structure of a database and aid in the communication between stakeholders, designers, and developers.

ERDs are considered a vital component in the analysis and design phase of database development (Elmasri & Navathe, 2015). These diagrams assist in identifying entity relationships, cardinalities, and constraints, enabling efficient database implementation and maintenance.

ER Notation

ER Notation, also known as Entity Relationship Notation, is a standardized graphical representation used in Entity Relationship Diagrams (ERDs) to depict the entities, attributes, and relationships within a database system. ER Notation employs a set of symbols and conventions that help in illustrating the structure and behavior of the database entities and their associations.

- **Entities:** Represent real-world objects or concepts within the system and are depicted as rectangles (Kroenke & Auer, 2016).
- **Attributes:** Represent the characteristics or properties of entities and are shown as ovals connected to the respective entity rectangle (Elmasri & Navathe, 2015).
- **Relationships:** Indicate the associations and dependencies between entities and are represented by diamond-shaped symbols (Kroenke & Auer, 2016)
- **Cardinality:** Describes the number of instances of one entity that are associated with another entity.
- **Primary Keys:** Denoted by underlined attributes, they represent unique identifiers for entities.
- **Foreign Keys:** Represented by dashed underline, they establish associations between entities.

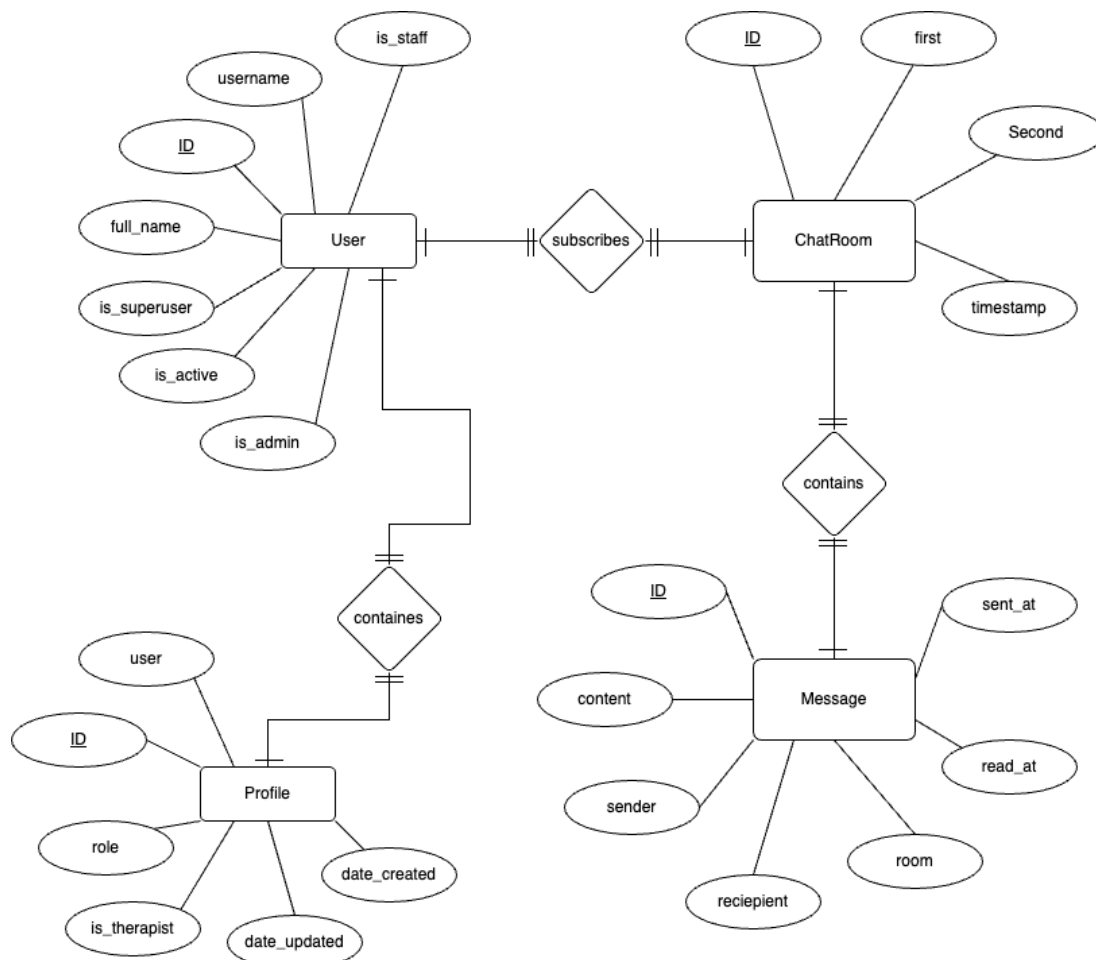


FIGURE 8: ENTITY RELATIONSHIP DIAGRAM (ERD)

(Source: Researcher, 2023)

4.7.6. Physical Data Model

The physical data model is a representation of the database design at the physical level, focusing on the implementation details of the database system. It describes how the logical data model, which includes entities, attributes, and relationships, is translated into the specific database management system (DBMS) and storage structures (Connolly & Begg, 2014).

In the physical data model, the entities and their attributes are mapped to the corresponding database tables and columns, taking into consideration data types, constraints, and indexing requirements. The relationships between entities are represented using primary and foreign keys to establish referential integrity and enforce data integrity rules.

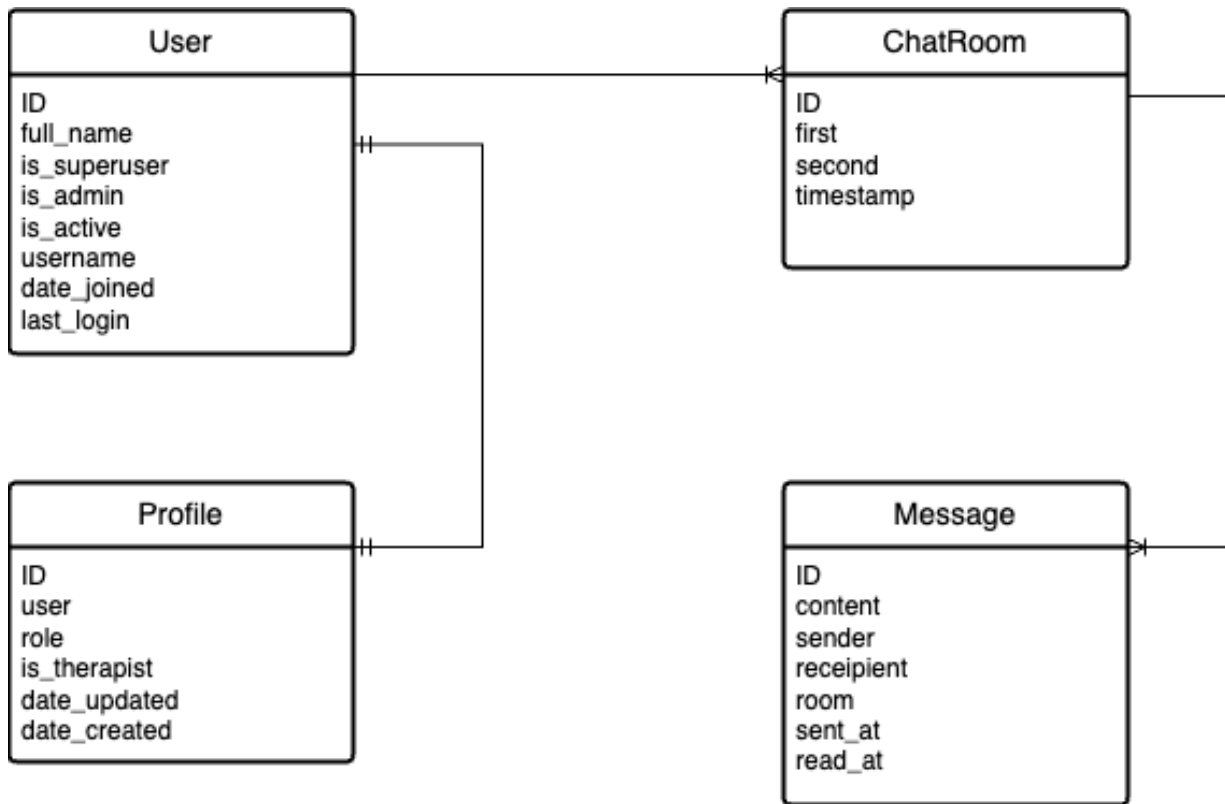


FIGURE 9: PHYSICAL DATA MODEL

(Source: Researcher, 2023)

4.8. Architecture of the System

The architecture of the mental health platform is designed to provide a robust and scalable system that effectively addresses the needs of users seeking mental health support. The architecture incorporates various components and technologies to ensure efficient data processing, seamless communication, and secure user interactions.

The system follows a client-server model, where clients interact with the platform through web or mobile interfaces, and the server handles the core functionalities and data processing. The architecture consists of the following components:

4.8.1. Presentation Layer:

This layer includes the user interfaces that allow users to access the platform. It comprises web pages or mobile app screens designed for easy navigation and interaction.

4.8.2. Application Layer

The application layer consists of the business logic and functionality of the mental health platform. It handles user requests, processes data, and performs various operations such as user authentication, appointment scheduling, and data retrieval.

4.8.3. Database Layer

The database layer stores and manages the persistent data of the system. It includes the user profiles, appointment records, chat logs, and other relevant information. This application uses Redis which acts as a message broker by helping users communicate by sending messages through Redis. With Django Channels, Redis plays a crucial role in facilitating real-time communication between the Django server and connected clients, ensuring efficient and scalable handling of requests and updates.

4.8.4. Communication Layer

The communication layer enables real-time communication between users and mental health professionals. Websockets technology is employed to facilitate instant messaging, allowing users to engage in interactive and timely conversations with professionals (Mozilla Developer Network, n.d.).

4.8.5. Security Layer

The security layer ensures the confidentiality, integrity, and privacy of user data. It incorporates measures such as encryption, access controls, and user authentication mechanisms to protect sensitive information and maintain trust in the platform.

4.8.6. Scalability and Performance

The architecture is designed to handle a large number of users and concurrent interactions. It employs load balancing techniques, caching mechanisms, and scalable infrastructure to ensure optimal performance and responsiveness.

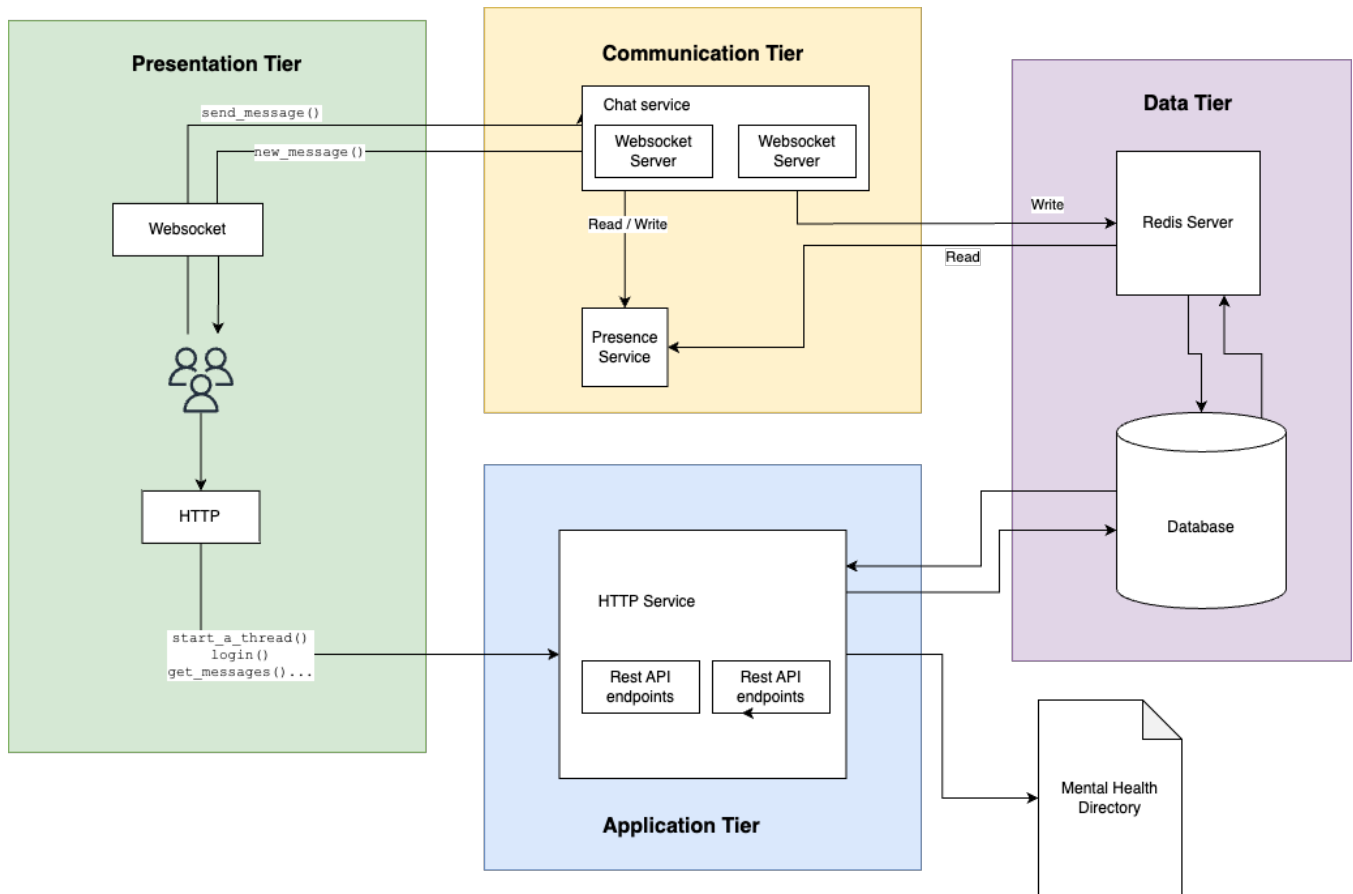


FIGURE 10: APPLICATION ARCHITECTURE

(Source: Researcher, 2023)

4.9. Implementation and coding

4.9.1. Introduction

This area contains the tools that were used by the researcher while developing the system, it also showcases the screenshots of the system, the source codes and the testing carried out.

4.9.2. Description of implementation Tools and technology

The tools mentioned below were used by the researcher in the development of the system.

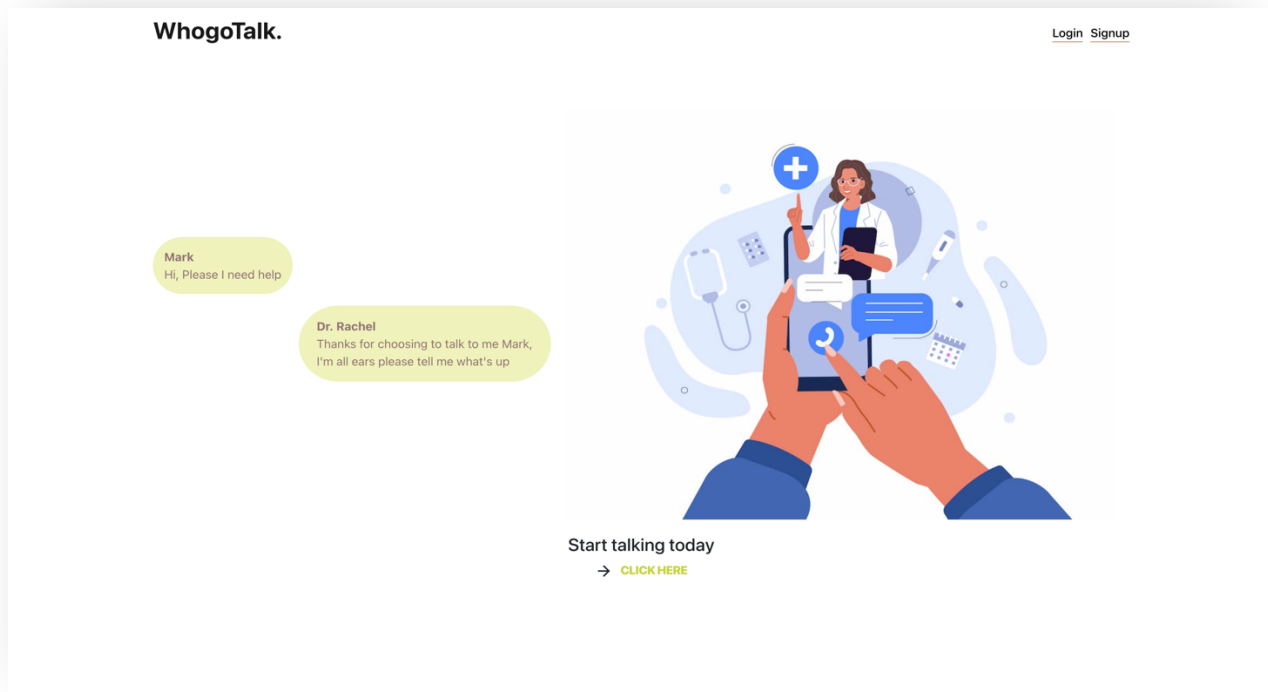
- Visual Studio Code (VS Code)
- React Developer Tools
- Node JS

Technologies used in developing the system

- Frontend Technologies: ReactJS, JSX, Bootstrap 5.2, CSS3
- Backend Technologies: Django, Django Rest Framework, Django Channels, Redis Server, ASGI
- Database: SQLite, PostgreSQL

4.9.3. Screenshots and source code

Landing Page



(Source: Researcher, 2023)

Source Code

```
import React, { useEffect } from "react";
import "../styles/Landing.css";
import ChatImage from "../images/talk2.jpg";
import ArrowForwardIcon from "@material-ui/icons/ArrowForward";
import { Link } from "react-router-dom";
import axios, { baseURL } from "../axios/axios";

const LandingPage = () => {
  useEffect(() => {
    // wakeup the Backend App because it takes 15s for heroku app to wakeup
    axios({
      method: "get",
      url: `${baseURL}/`,
    });
  });
}
```

```

    });
  }, []);
  return (
    <div className="landing-page-container container">
      <div className="landingPage">
        <div className="anime_chat_wrap">
          <p className="anime_chat text-start chat1">
            <span className="fw-bold">Mark</span> <br />
            Hi, Please I need help
          </p>
          <p className="anime_chat text-start py-3 px-4 chat2 move__right">
            <span className="fw-bold">Dr. Rachel</span>{" "}
            <br />
            Thanks for choosing to talk to me Mark, <br />{" "}
            I'm all ears please tell me what's up
          </p>
          <br />
        </div>
        <img src={ChatImage} alt="landing_image" />
      </div>

      <div className="landing__call">
        <h4>Start talking today</h4>
        <span className="landing__call_btns">
          <ArrowForwardIcon />
          <Link className="call_anc" to="/login">
            CLICK HERE
          </Link>
        </span>
      </div>
    </div>
  );
};

export default LandingPage;

```

Login Page

Email:

Password

login

Dont have an account yet? [Signup](#)

Source Code

```
import React, { useEffect } from "react";
import { connect } from "react-redux";
import "../styles/Login.css";
import { Link, useLocation } from "react-router-dom";
import { useFormik } from "formik";
import { loginFormvalidate } from "../utils/formValidation";
import { loginUserAction } from "../Redux/actions/authActions";
import { closeAlert } from "../Redux/actions/genericActions";

const Login = ({
  loginUserAction,
  show_alert,
  alert_message,
  alert_level,
  closeAlert,
  user,
  authenticated,
}) => {
  const formik = useFormik({
    initialValues: {
      email: "",
      password: "",
    },
    validate: loginFormvalidate,

    onSubmit: (values) => {
      console.log(values);
      loginUserAction(values);
    },
  });

  let location = useLocation();
  useEffect(() => {
    closeAlert();
  });
}
```

```

    }, [location]);

    return (
      <div className="login">
        <div className="login__box">
          <form onSubmit={formik.handleSubmit}>
            <div className="input__wrap">
              <label>Email:</label>
              <input
                id="email"
                name="email"
                onChange={formik.handleChange}
                value={formik.values.email}
                type="email"
              />
              {formik.errors.email ? (
                <div className="error">
                  {formik.errors.email}
                </div>
              ) : null}
            </div>
            <div className="input__wrap">
              <label htmlFor="password">Password</label>
              <input
                id="password"
                name="password"
                onChange={formik.handleChange}
                value={formik.values.password}
                type="password"
              />
              {formik.errors.password ? (
                <div className="error">
                  {formik.errors.password}
                </div>
              ) : null}
            </div>
            <div className="input__wrap">
              <button type="submit" className="submit__btn">
                login
              </button>
            </div>

            <i>
              Dont have an account yet?
              <Link to="/signup"> Signup</Link>
              {authenticated && (
                <p>Hello i am authenticated</p>
              )}
            </i>
          </form>
        </div>
      </div>
    );
  };

  function mapStateToProps(store) {
    return {
      show_alert: store.generic.show_alert,
      alert_message: store.generic.alert_message,
      alert_level: store.generic.alert_level,
      user: store.auth.user,
      authenticated: store.auth.is_auth,
    };
  }

```

```
};  
}  
  
export default connect(mapStateToProps, {  
  loginUserAction,  
  closeAlert,  
})(Login);
```

Admin Panel

Site administration

ACCOUNTS		
Users	+ Add	✎ Change
AUTH TOKEN		
Tokens	+ Add	✎ Change
AUTHENTICATION AND AUTHORIZATION		
Groups	+ Add	✎ Change
CHAT		
Chat rooms	+ Add	✎ Change
Messages	+ Add	✎ Change

Recent actions

My actions

 salam@gmail.com
 User

 williams23@gmail.com
 User

 salam@gmail.com
 User

 williams23@gmail.com
 User

 ngeyennv@gmail.com
 User

 williams23@gmail.com
 User

 williams23@gmail.com
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 User

 williams23@gmail.com
 User

 williams23@gmail.com
 User

Source Code

```
{% load i18n static %}
<!DOCTYPE html>
{% get_current_language as LANGUAGE_CODE %}{% get_current_language_bidi as LANGUAGE_BIDI %}
<html lang="{{ LANGUAGE_CODE|default:"en-us" }}" {% if LANGUAGE_BIDI %}dir="rtl"{% endif %}>
<head>
<title>{% block title %}{% endblock %}</title>
<link rel="stylesheet" type="text/css" href="{% block stylesheet %}{% static "admin/css/base.css" %}{%
endblock %}">
{% block extrastyle %}{% endblock %}
{% if LANGUAGE_BIDI %}<link rel="stylesheet" type="text/css" href="{% block stylesheet_rtl %}{% static
"admin/css/rtl.css" %}{% endblock %}">{% endif %}
{% block extrahead %}{% endblock %}
{% block responsive %}
    <meta name="viewport" content="user-scalable=no, width=device-width, initial-scale=1.0, maximum-
scale=1.0">
    <link rel="stylesheet" type="text/css" href="{% static "admin/css/responsive.css" %}">
    {% if LANGUAGE_BIDI %}<link rel="stylesheet" type="text/css" href="{% static
"admin/css/responsive rtl.css" %}">{% endif %}
{% endblock %}
```

```

{% endblock %}
{% block blockbots %}<meta name="robots" content="NONE,NOARCHIVE">{% endblock %}
</head>
{% load i18n %}

<body class="{% if is_popup %}popup {% endif %}{% block bodyclass %}{% endblock %}"
    data-admin-utc-offset="{% now "Z" %}">

<!-- Container -->
<div id="container">

    {% if not is_popup %}
    <!-- Header -->
    <div id="header">
        <div id="branding">
            {% block branding %}{% endblock %}
        </div>
        {% block usertools %}
        {% if has_permission %}
        <div id="user-tools">
            {% block welcome-msg %}
            {% trans 'Welcome,' %}
            <strong>{% firstof user.get_short_name user.get_username %}</strong>.
            {% endblock %}
            {% block userlinks %}
            {% if site_url %}
            <a href="{{ site_url }}">{% trans 'View site' %}</a> /
            {% endif %}
            {% if user.is_active and user.is_staff %}
            {% url 'django-admindocs-docroot' as docsroot %}
            {% if docsroot %}
            <a href="{{ docsroot }}">{% trans 'Documentation' %}</a> /
            {% endif %}
            {% endif %}
            {% if user.has_usable_password %}
            <a href="{% url 'admin:password_change' %}">{% trans 'Change password' %}</a> /
            {% endif %}
            <a href="{% url 'admin:logout' %}">{% trans 'Log out' %}</a>
            {% endblock %}
        </div>
        {% endif %}
    {% endblock %}
    {% block nav-global %}{% endblock %}
</div>
<!-- END Header -->
{% block breadcrumbs %}
<div class="breadcrumbs">
<a href="{% url 'admin:index' %}">{% trans 'Home' %}</a>
{% if title %} &rsaquo; {{ title }}{% endif %}
</div>
{% endblock %}
{% endif %}

{% block messages %}
{% if messages %}
<ul class="messagelist">{% for message in messages %}
    <li{% if message.tags %} class="{{ message.tags }}"{% endif %}>{{ message|capfirst }}</li>
{% endfor %}</ul>
{% endif %}
{% endblock messages %}

<!-- Content -->

```

```

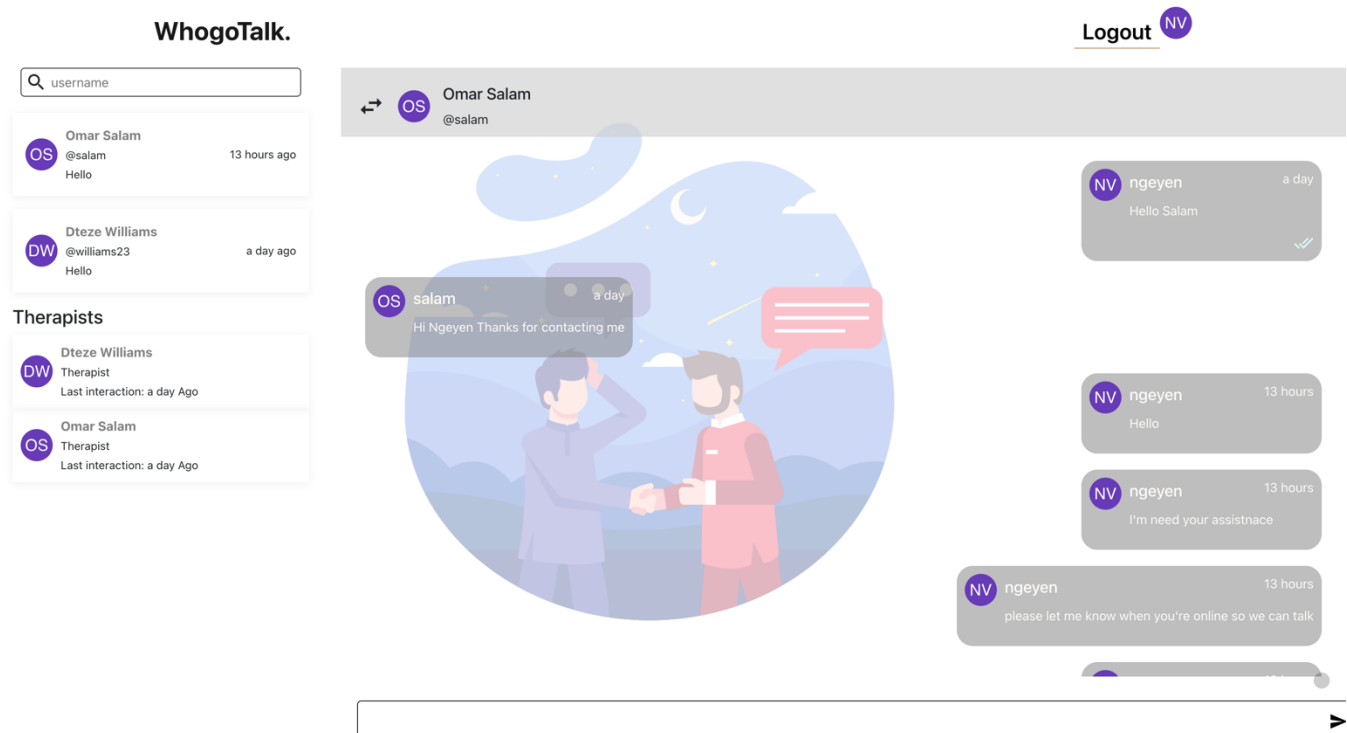
<div id="content" class="{% block coltype %}colM{% endblock %}">
  {% block pretitle %}{% endblock %}
  {% block content_title %}{% if title %}<h1>{{ title }}</h1>{% endif %}{% endblock %}
  {% block content %}
  {% block object-tools %}{% endblock %}
  {{ content }}
  {% endblock %}
  {% block sidebar %}{% endblock %}
  <br class="clear">
</div>
<!-- END Content -->

  {% block footer %}<div id="footer"></div>{% endblock %}
</div>
<!-- END Container -->

</body>
</html>

```

Chat Page



Source Code

```

import React, { useState, useEffect } from "react";
import { connect } from "react-redux";
import "../styles/chat/ChatArea.css";
import ChatMain from "../ChatMain";
import ChatSidebar from "../ChatSidebar";
import { setActiveChat } from "../../Redux/actions/chatActions";
import {
  connectToSocketAction,

```

```

    sendNewMessageAction,
  } from "../../Redux/actions/chatSocketActions";
import ChatContacts from "../ChatContacts";
import { Height } from "@material-ui/icons";

const ChatArea = ({
  isActiveChat,
  connectToSocketAction,
}) => {
  const [width, setWidth] = useState(window.innerWidth);
  const [openMain, setOpenMain] = useState(true);
  const [openSide, setOpenSide] = useState(true);

  useEffect(() => {
    connectToSocketAction();
  });

  window.addEventListener("resize", function () {
    setWidth(window.innerWidth);
  });

  useEffect(() => {
    if (width > 768) {
      setOpenMain(true);
      setOpenSide(true);
    } else {
      if (isActiveChat) {
        setOpenSide(false);
        setOpenMain(true);
      } else {
        setOpenMain(false);
        setOpenSide(true);
      }
    }
  }, [width, isActiveChat]);

  return (
    <div className="row">
      <div className="col-md-6 col-lg-3 d-none d-md-block">
        <ChatSidebar className="col-lg-3" />
      </div>
      <div className="col-lg-9 col-md-6">
        <ChatMain />
      </div>
    </div>
  );
};

function mapStateToProps(store) {
  return {
    user: store.auth.user,
    isActiveChat: store.chat.is_active_conversation,
  };
}

export default connect(mapStateToProps, {
  setActiveChat,
  connectToSocketAction,
  sendNewMessageAction,
})(ChatArea);

```

4.10. Testing

4.10.1. Introduction

This section aims to provide an overview of the testing process conducted on the mental health platform. It outlines the objectives of testing and discusses various types of testing that were performed to ensure the system's reliability and functionality. The researcher emphasizes the importance of testing and presents the outcomes of unit testing, validation testing, integration testing, as well as functional and system testing. Additionally, the section highlights the tools utilized during unit testing and the benefits derived from the research process.

4.10.2. Objective of Testing

The basic goal of testing is to uncover as many flaws as possible in a software product while confirming whether the application meets the user's needs (Pressman, 2014). Defects should be found as early in the testing cycle as feasible. Software testing helps in finalizing the software application or product against business and user requirements. It is very important to have good test coverage in order to test the software application completely and make sure that it is performing well and as per the specifications (Beizer, 1990). While determining the test coverage, the test cases should be well-designed with maximum possibilities of finding errors or bugs. The effectiveness of the test cases can be measured by the number of defects reported per test case, where a higher number indicates more effective test cases. (Kaner, et al., 1999).

4.10.3. Unit Testing Outputs

Unit testing is a critical aspect of software development that focuses on testing individual units or components of the software in isolation. The primary goal of unit testing is to verify the correctness and reliability of these units, ensuring that they function as intended and meet the specified requirements.

One of the key outputs of unit testing is the detection and identification of defects or errors within the units. By subjecting each unit to a battery of test cases, developers can identify any issues that may lead to incorrect behavior or failure. These defects can range from logical errors, boundary cases, to edge cases that were not initially considered. By catching these defects early in the development process, developers can address them promptly and minimize their impact on the overall system.

Regarding our mental health platform, unit testing was carried out following a well-structured plan. The initial code and functionalities were developed, and comprehensive tests were designed to evaluate the

individual components thoroughly. The focus of the unit tests was on critical features like user registration, login process, and data retrieval to ensure their accuracy and reliability in supporting mental health users.

The reasons for conducting unit testing in the context of our mental health platform are as follows:

- **Early identification of issues:** Unit tests help in detecting potential flaws and bugs at an early stage of development. By catching these issues early, we can address them promptly and prevent them from escalating into more complex problems, thereby saving time and resources.
- **Improved code understanding and maintenance:** Unit tests serve as documentation of the system's functionalities and expected behaviors. Developers can refer to these tests to understand how individual components should work, making it easier to maintain and modify the code in the future.
- **Ensuring data accuracy and privacy:** Unit tests verify the proper functioning of data retrieval and storage mechanisms, ensuring that user information remains secure and confidential. This is crucial for a mental health platform that deals with sensitive user data.
- **Code reliability and reusability:** By developing robust unit tests, we ensure the reliability of our code. Additionally, these tests can be reused in subsequent iterations or versions of the platform, enabling us to make changes and enhancements confidently without compromising existing functionality.

Unit Testing Tools

In the development and testing of our mental health platform, several unit testing tools were utilized to ensure the reliability and quality of the code. Unit testing in the context of Django, Django Rest Framework, Redis, JavaScript, and JSX, there are several tools available that can aid in the testing process. These tools provide specific functionalities and features tailored for each technology. Here are some commonly used unit testing tools for these frameworks and languages:

- Django Testing Tools:
- **Django TestCase:** Django provides a built-in test framework with the `django.test.TestCase` class for writing unit tests for Django applications. It allows testing models, views, forms, and other components of the Django application.
- Django Rest Framework Testing Tools:

- **APITestCase:** Django Rest Framework (DRF) provides **APITestCase** for writing unit tests for API views and endpoints. It includes helper methods for making API requests and asserting responses.
- **Redis Unit Testing:**
- **pytest-redis:** For unit testing applications that use Redis, **pytest-redis** is a useful Python testing plugin that allows setting up and tearing down Redis instances in tests.
- **JavaScript and JSX Unit Testing Tools:**
- **Jest:** Jest is a popular JavaScript testing framework commonly used for testing React applications and components written in JSX. It provides powerful tools for snapshot testing, mocking, and coverage reporting.
- **React Testing Library:** This library is useful for testing React components, including those written in JSX. It encourages testing components in a way that resembles how users interact with the UI.

4.10.4. Validation Testing outputs

Validation testing is an important stage in the software testing process that focuses on evaluating the software system to ensure it meets the specified requirements and satisfies the user's needs. Validation testing for the mental health platform involves several stages to ensure its effectiveness and alignment with the intended purpose. These stages are tailored to validate the platform's functionalities and meet the specific needs of mental health support:

- **Requirement Validation:** This stage involves validating that the platform has been developed according to the specified requirements for mental health support. It ensures that the features and functionalities align with the goals of the platform, such as connecting users with doctors and volunteers, providing a knowledge repository, and facilitating live assistance.
- **User Acceptance Testing (UAT):** UAT is crucial for the mental health platform to validate its usability and acceptance by the end users, including individuals seeking mental health support. It involves conducting tests and gathering feedback from users to ensure that the platform meets their needs, is user-friendly, and provides a positive user experience.
- **Functional Testing:** Functional testing focuses on validating the individual features and functionalities of the mental health platform. It includes testing components such as user registration, search and matching algorithms, messaging and communication systems, access to

the knowledge repository, and user support features. This stage ensures that each function performs as intended and delivers the desired outcomes.

- **Performance Testing:** Performance testing is essential to assess the platform's performance and responsiveness under various user loads. It includes testing factors such as response time, scalability, and system resource usage. This stage ensures that the platform can handle multiple concurrent users efficiently and provides a smooth and seamless experience.
- **Security Testing:** Security testing is crucial for the mental health platform to ensure the confidentiality, integrity, and privacy of user data. It involves testing authentication and authorization mechanisms, data encryption, secure communication protocols, and protection against potential vulnerabilities or threats. This stage ensures that the platform safeguards sensitive information and maintains a secure environment for users.
- **Compatibility Testing:** Compatibility testing is performed to ensure that the mental health platform functions correctly across different devices, browsers, and operating systems. It ensures that users can access and use the platform seamlessly regardless of their preferred devices or platforms, enhancing accessibility and usability.
- **Regression Testing:** Regression testing is conducted after making any updates or changes to the mental health platform to ensure that existing functionalities are not adversely affected. It helps identify and address any unintended issues or regressions introduced during the development process, ensuring the stability and reliability of the platform.

4.10.5. Integration Testing Outputs

Integration Testing is defined as a type of testing where software modules are integrated logically and tested as a group. A typical software project consists of multiple software modules, coded by different programmers. The purpose of this level of testing is to expose defects in the interaction between these software modules when they are integrated. Integration Testing focuses on checking data communication amongst these modules.

Approaches, Strategies, Methodologies of Integration Testing

Software Engineering defines variety of strategies to execute Integration testing, viz.

- i. Big Bang Approach:
- ii. Incremental Approach: which is further divided into the following

- a. Top-Down Approach
- b. Bottom-Up Approach
- c. Sandwich Approach – Combination of Top Down and Bottom Up .

Whogotalk is built as an N-Tier Architecture with 4 tiers. The incremental approach to integrated testing adopted in the study allows for testing the platform's components in a phased manner, gradually integrating and testing new functionalities with existing ones. This approach helps identify and address any potential issues or conflicts that may arise during the integration process, ensuring that the platform functions smoothly and seamlessly as new features are added.

4.10.6. Functional and system testing Results.

Functional testing is a type of testing that seeks to establish whether each application feature works as per the software requirements. Each function is compared to the corresponding requirement to ascertain whether its output is consistent with the end user's expectations. The testing is done by providing sample inputs, capturing resulting outputs, and verifying that actual outputs are the same as expected outputs.

The functional testing carried out included but not limited the following

- i) Can users successfully log in to the application once they provide legitimate credentials?
- ii) Can users send messages and communicate in real-time?
- iii) Can users see an unread messages counter?

The system was tested with respect to the functional tests stated above and the system passed these requirements, in other words, the system was able to do what it was set out to do in the functional requirements.

4.10.7. Acceptance Testing Report

Acceptance testing - is the final phase of product testing prior to public launch. Acceptance testing is performed once the product team has decided that the product is completely ready for the market. The User Acceptance Testing Report is the document produced to summarize testing efforts and evaluation results of the acceptance testing phase Below is the Acceptance Report.

N	Date	5 th September 2022
1	Test case execution status	▪ Number of test cases executed today: 10 ▪ Number of test cases passing: 10 ▪ Number of test cases failing 0

2	Defect identifier	- Submitted defect number: 0 - Brief description of issue: 0
3	ACC number(s)	Acceptance criteria change document number(s) if any
4	Cumulative test execution status	- Total number of test cases executed:20 - Total number of test cases passing: 17 - Total number of test cases failing: 3 - Total number of test cases not executed yet: 0

5. CHAPTER 5 CONCLUSIONS AND RECOMMENDATIONS

5.1. Introduction

In this chapter, the researcher looks at the conclusion and the recommendation regarding Online School bus management system.

5.2. Conclusion

In conclusion, this study has examined the development and potential implementation of a mental health platform to address the existing gaps in accessing mental health services in the African context. Through a thorough literature review, it has been established that mental health issues are often stigmatized in many African communities, resulting in limited access to services and inadequate support for individuals in need. The existing platforms and initiatives in the field have provided valuable insights into the challenges and opportunities in this area.

The proposed mental health platform aims to bridge the gap between the public and mental health professionals by providing a secure and user-friendly platform for individuals to seek support and connect with licensed therapists and trained volunteers. By utilizing technology, this platform can overcome geographical barriers and provide access to mental health services, particularly in underserved areas. Furthermore, it can help address the stigma associated with mental health topics by offering a confidential and non-judgmental space for individuals to seek help.

Based on the responses gathered from the participants, it was found that a significant percentage of respondents rated their mental health as average, while a notable proportion reported poor mental health. This highlights the need for accessible mental health services for these communities, as they are often marginalized and face social exclusion due to their mental health issues.

The implementation of unit testing and validation testing has ensured the quality and reliability of the system. The use of appropriate testing tools and methodologies has helped identify and address bugs and errors, leading to an improved user experience and system performance.

The study was guided also with specific objectives:

- Develop a user-friendly platform for mental health education and support.
- Create a secure login and registration system for users, doctors, and volunteers.
- Allow users to search for and connect with volunteer mental health experts for live chat sessions.
- Enable users to search for mental health information through an integrated knowledge repository.

- Provide an appointment scheduling system for users to book virtual consultations with mental health professionals.

In conclusion, the study achieved its general objective that was stated out, which was to create a platform that connects individuals in African communities with licensed therapists and trained volunteers.

5.3. Recommendations

The mental health platform demonstrates efficiency and reliability in addressing the needs of underserved communities. Based on the findings and analysis, the following specific recommendations are made:

1. **Outreach and Awareness:** Conduct extensive outreach and awareness campaigns to ensure maximum reach and adoption of the platform. Collaborate with local organizations, schools, and healthcare providers to promote the platform and its benefits. This will help in creating a strong user base and fostering a sense of trust and credibility.
2. **Enhanced User Engagement:** Improve user engagement by incorporating additional features such as interactive forums, virtual support groups, and personalized user profiles. These features will encourage active participation and create a sense of community among users. Consider incorporating gamification elements to motivate users and reward their engagement.
3. **Continuous Improvement:** Regularly evaluate and enhance the platform based on user feedback, surveys, and analytics. Continuously update the platform's content, features, and design to meet the evolving needs of the target population. Stay up-to-date with emerging research and best practices in the field of mental health to ensure the platform remains effective and relevant.
4. **Collaborate with Mental Health Professionals:** Establish partnerships with mental health professionals, therapists, and counselors to provide expert guidance and support on the platform. Enable users to connect with professionals through secure communication channels or virtual consultations. This collaboration will enhance the platform's credibility and ensure the provision of evidence-based interventions.
5. **Scalability and Security:** Design the platform with scalability in mind to accommodate a growing user base and increasing demand for services. Implement robust security measures to protect user data and ensure compliance with data protection regulations. Regularly conduct security audits and vulnerability assessments to identify and address any potential risks or breaches.
6. **Mobile App Development:** Consider developing a mobile app version of the mental health platform to cater to users who prefer mobile devices. This will enhance accessibility and convenience, allowing users to access the platform on-the-go and receive timely support.

By implementing these recommendations, the mental health platform can effectively address the mental health needs of underserved communities, improve accessibility to support services, and contribute to the overall well-being of individuals in these communities.

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APPENDICES

TIME FRAME

Stages	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sept	Oct	Nov	Dec
Identification of Topic												
Review Collection												
Writing Proposal												
Presentation of Proposal												
Development of Instruments												
Administration of Questionnaire												
Data Analysis & Discussion of findings												
System Development and Testing												
Report writing and Final report Submission												

QUESTIONNAIRE

Hello, I am Ngeyen Victory, a student at the University of Kigali pursuing a bachelor's degree in Business Computer Science. I am finalizing my studies at the University, and I am currently writing my dissertation. My dissertation topic is titled **A Mental Health Platform for African Communities** and would like to request you to take time and fill out this questionnaire that will facilitate me to further proceed with my research writing. The responses will anonymous and will only be used for research purposes. Furthermore, the responses will be treated with utmost confidentiality.

- What's your Sex?
 - ☐ Male
 - ☐ Female
- What's your Age group?
 - ☐ Between 13 above 18
 - ☐ Between 19 and 25
 - ☐ Between 26 to 39
 - ☐ 40 and above
- What is your relationship status?
 - ☐ Single
 - ☐ Married
 - ☐ Widowed
 - ☐ Divorced
 - ☐ Separated
- Overall how would you rate your mental health?
 - ☐ Excellent
 - ☐ Somewhat good
 - ☐ Average
 - ☐ Somewhat poor
 - ☐ Poor
 - ☐ Not sure

- During the past 4 weeks, have you had any problems with your work or daily life due to any emotional problems, such as feeling depressed, sad or anxious?
 - ☐ Yes
 - ☐ No
 - ☐ Not sure
- During the past 4 weeks, how often has your mental health affected your ability to get work done?
 - ☐ Very often
 - ☐ Somewhat often
 - ☐ Not so often
 - ☐ Not at all
- Have you felt particularly low or down for more than 2 weeks in a row?
 - ☐ Very often
 - ☐ Somewhat often
 - ☐ Not so often
 - ☐ Not at all
- During the past two weeks, how often has your mental health affected your relationships?
 - ☐ Very often
 - ☐ Somewhat often
 - ☐ Not so often
 - ☐ Not at all
- How often do you experience below?

	Never	Once in a while	About half the time	Most of the time	Always
Calm and peaceful					
Energetic					
Gloomy					
Angry					

- Have you noticed any change in your diet habits?
 - ☐ Yes, I eat too much

- ☐ Yes, I don't feel hungry
- ☐ Not much
- ☐ No change
- When was the last time you were really happy?
 - ☐ Few days ago
 - ☐ Few weeks ago
 - ☐ Few months ago
 - ☐ Few years ago
 - ☐ I don't remember
- When was the last time you felt good about yourself?
 - ☐ Few days ago
 - ☐ Few weeks ago
 - ☐ Few months ago
 - ☐ Few years ago
 - ☐ I don't remember
- Have you ever been diagnosed with a mental disorder before?
 - ☐ Yes
 - ☐ No
 - ☐ Not sure
- When did you last get your mental health examination done?
 - ☐ Less than 6 months ago
 - ☐ 6 months ago
 - ☐ A year ago
 - ☐ More than a year ago
 - ☐ Never
- Have you seen a therapist in the recent past?
 - ☐ Yes
 - ☐ No

THANK YOU